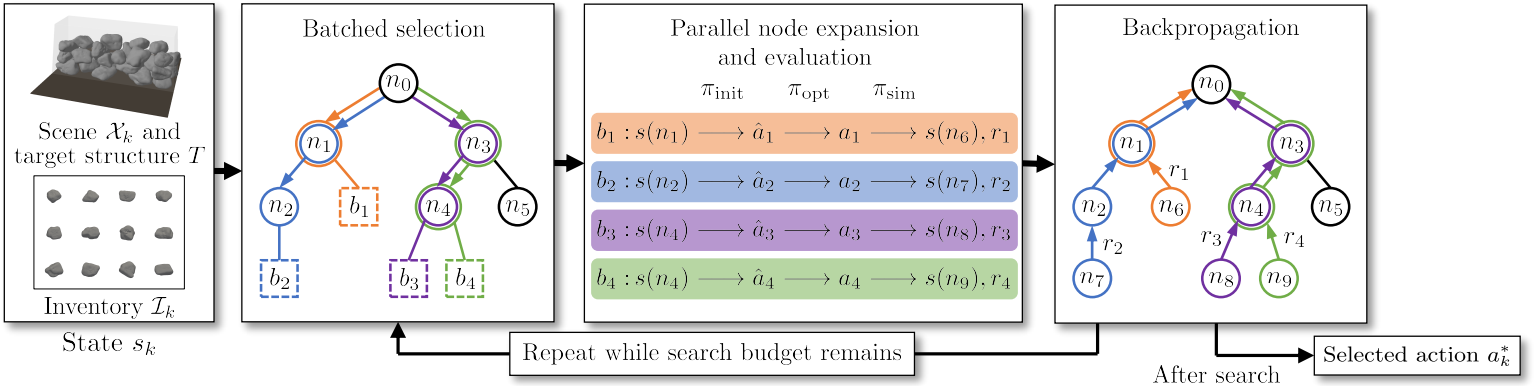

Scene $\mathcal{X}_k$ and target structure $T$
Inventory $\mathcal{I}_k$
State $s_k$
Batched selection
$n_0$
$n_1$
$n_2$
$b_1$
$n_3$
$n_4$
$n_5$
$b_2$
$b_3$
$b_4$
Parallel node expansion and evaluation
$\pi_{\text{init}}$ $\pi_{\text{opt}}$ $\pi_{\text{sim}}$
$b_1 : s(n_1) \longrightarrow \hat{a}_1 \longrightarrow a_1 \longrightarrow s(n_6), r_1$
$b_2 : s(n_2) \longrightarrow \hat{a}_2 \longrightarrow a_2 \longrightarrow s(n_7), r_2$
$b_3 : s(n_4) \longrightarrow \hat{a}_3 \longrightarrow a_3 \longrightarrow s(n_8), r_3$
$b_4 : s(n_4) \longrightarrow \hat{a}_4 \longrightarrow a_4 \longrightarrow s(n_9), r_4$
Backpropagation
$n_0$
$n_1$
$n_2$
$n_6$
$r_1$
$n_3$
$n_4$
$n_5$
$r_2$
$r_3$
$r_4$
$n_7$
$n_8$
$n_9$
Repeat while search budget remains
After search
Selected action $a_k^*$